Topic I: Principal Component Analysis

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Before getting into the details of principal component analysis (PCA), we discuss its two main uses, which are related.

1. Dimensionality reduction
2. Data compression

Note that these terms are not being used or defined precisely here – they are informal notions, but useful ones nevertheless. Loosely speaking, "dimensionality reduction" refers to embedding a data set $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ into a low-dimensional space $\mathbb{R}^d$, where $d \cdot n$ is much smaller than the size of the original dataset, and where much of the "important structure" of the data is preserved in the embedded space. For example, distances between the data vectors may be (approximately) preserved by the embedding.

"Data compression" is similar, and refers to the process of finding a representation of the data that allows for reconstruction of the original data, with some loss. For example, if $\mathbf{x}_1, \dots, \mathbf{x}_n$ are vectors in $\mathbb{R}^p$ for a very large p, we would like to find a small number of vectors $\mathbf{u}_1, \dots, \mathbf{u}_d \in \mathbb{R}^p$ (with $d \ll p$) and coefficients β_{ij} such that $\mathbf{x}_i \approx \beta_{i1}\mathbf{u}_1 + \cdots + \beta_{id}\mathbf{u}_d$; this way, we can store the data (with some small error) at storage cost $O((p+n)\cdot d)$, rather than $O(p \cdot n)$.

Speaking again at a very high level, the main difference between dimensionality reduction and data compression is that with data compression, one can (approximately) recover the original data from its compressed form. With dimensionality reduction this recovery is not always possible, but because the important structure of the data is preserved in the low-dimensional space it may not be necessary to work with the original data.

Principal component analysis is a classical method (dating back to a 1901 work by Karl Pearson [5], and developed independently in the 1930's by Hotelling [2]) for achieving these two tasks. Despite its age, however, it is still widely used and is still an active topic of research. Though we will focus only on certain aspects of PCA, more comprehensive treatments can be found in, for example, the references [3] and [4].

There is another reason for covering PCA in detail, namely that it gives us an excuse to learn fundamental results from linear algebra that may not have been covered in an introductory linear algebra course. In particular, we will give a detailed proof of (a version of) the Davis-Kahan $\sin\Theta$ Theorem, whose usefulness permeates applied mathematics; doing so will entail learning many other useful results along the way.

1 Principal components and principal values

Let $X \in \mathbb{R}^p$ be a random vector. Denote its mean vector by $\boldsymbol{\mu}$

$$\boldsymbol{\mu} = \mathbb{E}[X], \tag{1}$$

and its covariance matrix by

$$\boldsymbol{\Sigma} = \mathbb{E}[(X-\boldsymbol{\mu})(X-\boldsymbol{\mu})^T]. \tag{2}$$

For any deterministic unit vector $\mathbf{u} \in \mathbb{R}^p$, we define the variance of X along $\mathbf{u}$ to be

$$\mathrm{Var}[X^T\mathbf{u}], \tag{3}$$

i.e. the variance of the random variable $X^T\mathbf{u}$.

The first principal component of X, call it $\mathbf{u}_1$, is defined via

$$\mathbf{u}_1 \in \operatorname*{argmax}_{\mathbf{u}:\|\mathbf{u}\|=1} \mathrm{Var}[X^T\mathbf{u}], \tag{4}$$

where the symbol $\in$ here indicates that we take any vector which achieves the maximum. That is, $\mathbf{u}_1$ is a direction of maximum variance of X. The first principal value of X, which we will denote by σ_1^2, is then defined by

$$\sigma_1^2 = \mathrm{Var}[X^T \mathbf{u}_1]. \tag{5}$$

We now recursively define the principal components and values of X. For $k \geq 2$, the k-th principal component $\mathbf{u}_k$ is given by

$$\mathbf{u}_k \in \operatorname*{argmax}_{\mathbf{u}:\|\mathbf{u}\|=1, \mathbf{u}\perp\mathbf{u}_1,\dots,\mathbf{u}_{k-1}} \mathrm{Var}[X^T \mathbf{u}], \tag{6}$$

and the corresponding principal value σ_k^2 is defined by

$$\sigma_k^2 = \mathrm{Var}[X^T \mathbf{u}_k]. \tag{7}$$

2 Principal components and the covariance matrix

How do we actually compute the principal components and principal values of a random vector X? We will show that the principal components are the eigenvectors of the covariance matrix $\mathbf{\Sigma}$ of X, and the principal values are the corresponding eigenvalues. We first recall the spectral theorem from linear algebra along with the variational formulation of eigenvectors and eigenvalues.

Theorem 2.1 (Spectral Theorem). *If $\mathbf{A} \in \mathbb{R}^{p\times p}$ is a real, symmetric matrix, then it has an orthonormal basis of eigenvectors $\mathbf{w}_1, \dots, \mathbf{w}_p$ and real eigenvalues $\lambda_1, \dots, \lambda_p$.*

Theorem 2.2 (Rayleigh-Ritz). *If $\mathbf{A} \in \mathbb{R}^{p\times p}$ is a real, symmetric matrix with an orthonormal basis of eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_p$ and real eigenvalues $\lambda_1 \geq \cdots \geq \lambda_p$, then*

$$\mathbf{u}_k \in \operatorname*{argmax}_{\substack{\|\mathbf{w}\|=1: \\ \mathbf{w}\perp\mathbf{u}_1,\dots,\mathbf{u}_{k-1}}} \mathbf{w}^T \mathbf{A} \mathbf{w} \tag{8}$$

and

$$\lambda_k = \max_{\substack{\|\mathbf{w}\|=1: \\ \mathbf{w}\perp\mathbf{w}_1,\dots,\mathbf{w}_{k-1}}} \mathbf{w}^T \mathbf{A} \mathbf{w} = \mathbf{u}_k^T \mathbf{A} \mathbf{u}_k. \tag{9}$$

Proof. Suppose $\mathbf{A} = \sum_{j=1}^p \lambda_j \mathbf{u}_j \mathbf{u}_j^T$ is the eigenvalue decomposition of $\mathbf{A}$. Then for any unit vector $\mathbf{w}$

$$\mathbf{w}^T \mathbf{A} \mathbf{w} = \sum_{j=1}^p \lambda_j (\mathbf{w}^T \mathbf{u}_j)^2. \tag{10}$$

Since $\sum_{j=1}^p (\mathbf{w}^T \mathbf{u}_j)^2 = 1$, (10) is a convex combination of $\lambda_1, \dots, \lambda_p$, which is maximized by taking $(\mathbf{w}^T \mathbf{u}_1)^2 = 1$ and $(\mathbf{w}^T \mathbf{u}_j)^2 = 0$ for $j > 1$. This proves that $\mathbf{u}_1$ maximizes $\mathbf{w}^T \mathbf{A} \mathbf{w}$ over all unit vectors, with maximum value λ_1. We now proceed inductively; if $\mathbf{w}$ is orthogonal to $\mathbf{u}_1, \dots, \mathbf{u}_k$, then

$$\mathbf{w}^T \mathbf{A} \mathbf{w} = \sum_{j=k+1}^p \lambda_j (\mathbf{w}^T \mathbf{u}_j)^2, \tag{11}$$

and the sum is maximized by taking $(\mathbf{w}^T \mathbf{u}_{k+1})^2 = 1$ and $(\mathbf{w}^T \mathbf{u}_j)^2 = 0$ for $j > k+1$, proving that $\mathbf{u}_{k+1}$ maximizes $\mathbf{w}^T \mathbf{A} \mathbf{w}$ over all unit vectors orthogonal to $\mathbf{u}_1, \dots, \mathbf{u}_k$, with maximum value λ_{k+1}. $\square$

From these results, we show the following:

Proposition 2.3. *The PCs of X are the eigenvectors of $\mathbf{\Sigma}$, and the PVs are the corresponding eigenvalues.*

Proof. Let $\widetilde{X} = X - \boldsymbol{\mu}$ be the centered random vector. Then

$$\boldsymbol{\Sigma} = \mathbb{E}[\widetilde{X}\widetilde{X}^T]. \tag{12}$$

For any unit vector $\mathbf{u} \in \mathbb{R}^p$, the variance of X along $\mathbf{u}$ is then

$$\begin{aligned}
\mathrm{Var}[X^T\mathbf{u}] &= \mathrm{Var}[\widetilde{X}^T\mathbf{u}] \\
&= \mathbb{E}[(\widetilde{X}^T\mathbf{u})^2] \\
&= \mathbb{E}[(\widetilde{X}^T\mathbf{u})(\widetilde{X}^T\mathbf{u})] \\
&= \mathbb{E}[(\mathbf{u}^T\widetilde{X})(\widetilde{X}^T\mathbf{u})] \\
&= \mathbf{u}^T\mathbb{E}[\widetilde{X}\widetilde{X}^T]\mathbf{u}. \\
&= \mathbf{u}^T\boldsymbol{\Sigma}\mathbf{u}.
\end{aligned} \tag{13}$$

Consequently, the first PC, i.e. the unit vector $\mathbf{u}$ that maximizes $\mathrm{Var}[X^T\mathbf{u}]$, is the first eigenvector of $\boldsymbol{\Sigma}$ by Rayleigh-Ritz; the second PC is the second eigenvector of $\boldsymbol{\Sigma}$; and so on. □

Before continuing to applications of PCA, we state a result that is similar to in spirit to Rayleigh-Ritz, and which will prove quite useful later on, known as the Courant-Hilbert variational formula for eigenvalues:

Theorem 2.4 (Courant-Hilbert)**.** *Suppose* $\mathbf{A}$ *is a symmetric* n*-by-*n *matrix, with eigenvalues* $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$ *and corresponding orthonormal basis of eigenvectors* $\mathbf{u}_1, \dots, \mathbf{u}_n$*. Then:*

$$\lambda_k = \max_{\substack{\mathcal{H} \leq \mathbb{R}^n: \\ \dim \mathcal{H} = k}} \min_{\substack{\mathbf{v} \in \mathcal{H}: \\ \|\mathbf{v}\|=1}} \mathbf{v}^T\mathbf{A}\mathbf{v}. \tag{14}$$

Furthermore, if $\mathcal{H}_k = \mathrm{span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$*, then*

$$\lambda_k = \min_{\substack{\mathbf{v} \in \mathcal{H}_k: \\ \|\mathbf{v}\|=1}} \mathbf{v}^T\mathbf{A}\mathbf{v} = \mathbf{u}_k^T\mathbf{A}\mathbf{u}_k. \tag{15}$$

Proof of Courant-Hilbert. To show this, we work with the spectral decomposition of $\mathbf{A}$:

$$\mathbf{A} = \sum_{j=1}^n \lambda_j \mathbf{u}_j\mathbf{u}_j^T. \tag{16}$$

Take any $\mathcal{H} \leq \mathbb{R}^n$ with $\dim \mathcal{H} = k$. Then there must exist some $\mathbf{v}_\mathcal{H} \in \mathcal{H}$ with $\|\mathbf{v}_\mathcal{H}\| = 1$ and $\mathbf{v}_\mathcal{H} \perp \mathcal{H}_{k-1}$, by dimensionality considerations; specifically, since $\dim \mathcal{H} = k$ and $\dim \mathcal{H}_{k-1}^\perp = n - k + 1$, the dimensions of $\mathcal{H}$ and $\mathcal{H}_{k-1}^\perp$ add up to $n + 1 > n$, hence their intersection must have dimension at least 1. Since $\mathbf{v}_\mathcal{H}$ is orthogonal to $\mathbf{u}_1, \dots, \mathbf{u}_{k-1}$, we can write

$$\mathbf{v}_\mathcal{H} = \sum_{j=k}^n (\mathbf{v}_\mathcal{H}^T\mathbf{u}_j)\mathbf{u}_j, \tag{17}$$

(the sum starts at k since $\mathbf{v}_\mathcal{H} \perp \mathcal{H}_{k-1} = \mathrm{span}\{\mathbf{u}_1, \dots, \mathbf{u}_{k-1}\}$) and so

$$\min_{\substack{\mathbf{v} \in \mathcal{H}: \\ \|\mathbf{v}\|=1}} \mathbf{v}^T\mathbf{A}\mathbf{v} \leq \mathbf{v}_\mathcal{H}^T\mathbf{A}\mathbf{v}_\mathcal{H} = \sum_{j=k}^n \lambda_j(\mathbf{u}_j^T\mathbf{v}_\mathcal{H})^2 \leq \lambda_k \tag{18}$$

(the last bound holds since the values $(\mathbf{u}_k^T\mathbf{v}_\mathcal{H})^2, \dots, (\mathbf{u}_n^T\mathbf{v}_\mathcal{H})^2$ sum to 1, and $\lambda_k \geq \lambda_{k+1} \geq \dots \lambda_n$). Since (18) holds for all $\mathcal{H}$ with $\dim \mathcal{H} = k$, it shows that

$$\max_{\substack{\mathcal{H} \leq \mathbb{R}^n: \\ \dim \mathcal{H} = k}} \min_{\substack{\mathbf{v} \in \mathcal{H}: \\ \|\mathbf{v}\|=1}} \mathbf{v}^T\mathbf{A}\mathbf{v} \leq \lambda_k. \tag{19}$$

On the other hand we can achieve the upper bound λ_k by taking $\mathcal{H} = \mathcal{H}_k$ and $\mathbf{v} = \mathbf{u}_k$. Indeed, if $\mathbf{v} \in \mathcal{H}_k$, we have:

$$\mathbf{v}^T\mathbf{A}\mathbf{v} = \sum_{j=1}^k \lambda_j(\mathbf{u}_j^T\mathbf{v})^2, \tag{20}$$

which is minimized by taking $\mathbf{v} = \mathbf{u}_k$, in which case the sum is equal to λ_k; thereby proving that

$$\mathbf{u}_k^T \mathbf{A} \mathbf{u}_k = \min_{\substack{\mathbf{v} \in \mathcal{H}_k: \\ \|\mathbf{v}\|=1}} \mathbf{v}^T \mathbf{A} \mathbf{v} = \lambda_k. \tag{21}$$

This completes the proof. $\square$

3 PCA and dimensionality reduction

Let's assume we have a data set consisting of n vectors $\mathbf{x}_1, \dots, \mathbf{x}_n$. We can define the random vector X as taking each value $\mathbf{x}_i$ with probability $1/n$. Then the mean of X is where

$$\boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i, \tag{22}$$

and the covariance of X is

$$\boldsymbol{\Sigma} = \frac{1}{n} \sum_{i=1}^{n} (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^T. \tag{23}$$

We show that the PCs of X and mean $\boldsymbol{\mu}$ solve the *best subspace problem.* Fix $d \ge 1$, and consider the problem of approximating the vectors $\mathbf{x}_i$ by a d-dimensional affine subspace $\mathcal{A} = \mathbf{c} + \mathcal{U}$, where $\dim(\mathcal{U}) = d$ and $\mathbf{c} \in \mathbb{R}^p$. That is, we wish to find a matrix $\mathbf{U} \in \mathbb{R}^{p \times d}$ with orthonormal columns, a vector $\mathbf{c} \in \mathbb{R}^p$, and vectors $\boldsymbol{\beta}_i \in \mathbb{R}^d$ such that $\mathbf{x}_i \approx \mathbf{c} + \mathbf{U}\boldsymbol{\beta}_i$ for each i; more precisely, such that the error

$$\phi(\mathbf{c}, \mathbf{U}, \boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_n) = \sum_{i=1}^{n} \|\mathbf{c} + \mathbf{U}\boldsymbol{\beta}_i - \mathbf{x}_i\|^2 \tag{24}$$

is minimized.

We will prove:

Theorem 3.1 (Optimality of PCA)**.** *Let $\mathbf{u}_1, \dots, \mathbf{u}_p$ be the PCs of X, with PVs $\sigma_1^2 \ge \cdots \ge \sigma_p^2$. The functional $\phi(\mathbf{c}, \mathbf{U}, \boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_n)$ is minimized by taking $\mathbf{c} = \boldsymbol{\mu}$, $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_d]$, and $\beta_{ij} = \langle \mathbf{x}_i - \boldsymbol{\mu}, \mathbf{u}_j \rangle$, i.e. $\boldsymbol{\beta}_i = \mathbf{U}^T(\mathbf{x}_i - \boldsymbol{\mu})$. Furthermore,*

$$\frac{1}{n} \phi(\mathbf{c}, \mathbf{U}, \boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_n) = \sum_{k=d+1}^{p} \sigma_k^2. \tag{25}$$

We will make use of some basic facts from linear algebra, which we state without proof:

Lemma 3.2. *Let $\mathbf{A}$ and $\mathbf{B}$ be two matrices of the same size. Then $\langle \mathbf{A}, \mathbf{B} \rangle = \mathrm{tr}(\mathbf{A}^T \mathbf{B})$.*

Lemma 3.3. *Let $\mathbf{A}$ and $\mathbf{B}$ be two matrices such that $\mathbf{AB}$ and $\mathbf{BA}$ are well-defined (and square). Then $\mathrm{tr}(\mathbf{AB}) = \mathrm{tr}(\mathbf{BA})$.*

We will also need the following result:

Theorem 3.4 (Ky Fan)**.** *Let $\mathbf{A}$ be any real, symmetric p-by-p matrix with eigenvalues $\lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_p$, with $\mathbf{v}_1, \dots, \mathbf{v}_p$ a corresponding orthonormal basis of eigenvectors. Then*

$$\max_{\mathbf{V}^T\mathbf{V} = \mathbf{I}_d} \mathrm{tr}(\mathbf{V}^T \mathbf{A} \mathbf{V}) = \sum_{k=1}^{d} \lambda_k, \tag{26}$$

and the maximum is achieved by $\mathbf{V}_d = [\mathbf{v}_1, \dots, \mathbf{v}_d]$.

Proof. First, observe that if we can show

$$\max_{\mathbf{V}^T\mathbf{V} = \mathbf{I}_d} \mathrm{tr}(\mathbf{V}^T \mathbf{A} \mathbf{V}) = \sum_{k=1}^{d} \lambda_k, \tag{27}$$

then the maximum is achieved at $\mathbf{V}_d = [\mathbf{v}_1, \dots, \mathbf{v}_d]$. Indeed, by writing the eigendecomposition

$$\mathbf{A} = \sum_{k=1}^{p} \lambda_k \mathbf{v}_k \mathbf{v}_k^T, \tag{28}$$

we have, by the orthonormality of the eigenvectors of $\mathbf{A}$,

$$\mathbf{V}_d^T \mathbf{A} \mathbf{V}_d = \sum_{i=1}^{d} \sum_{j=1}^{d} \sum_{k=1}^{p} \lambda_k \mathbf{v}_i^T \mathbf{v}_k \mathbf{v}_k^T \mathbf{v}_j = \sum_{k=1}^{d} \lambda_k. \tag{29}$$

Now, we need to show (27). Let $\mathcal{S} = \{\mathbf{U}\mathbf{U}^T : \mathbf{U}^T\mathbf{U} = \mathbf{I}_d\}$, the set of all rank d orthogonal projection matrices. We want to find the maximum value of $\mathrm{tr}(\mathbf{U}^T\mathbf{A}\mathbf{U}) = \mathrm{tr}(\mathbf{A}\mathbf{U}\mathbf{U}^T)$, where $\mathbf{U}^T\mathbf{U} = \mathbf{I}_d$; or equivalently, maximize

$$f(\mathbf{W}) = \mathrm{tr}(\mathbf{A}\mathbf{W}) \tag{30}$$

over $\mathbf{W} \in \mathcal{S}$. Write the eigendecomposition of $\mathbf{A}$ as $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$, where $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_p]$. Then

$$f(\mathbf{W}) = \mathrm{tr}(\mathbf{A}\mathbf{W}) = \mathrm{tr}(\mathbf{\Lambda}\mathbf{V}^T\mathbf{W}\mathbf{V}). \tag{31}$$

Because $\mathbf{V}^T\mathbf{W}\mathbf{V}$ is in $\mathcal{S}$ if and only if $\mathbf{W}$ is in $\mathcal{S}$, we can without loss of generality assume $\mathbf{A} = \mathbf{\Lambda}$ is diagonal. With this, we observe

$$f(\mathbf{W}) = \mathrm{tr}(\mathbf{\Lambda}\mathbf{W}) = \sum_{i=1}^{p} \lambda_i w_{ii}. \tag{32}$$

We will need the following:

Lemma 3.5. *If $\mathbf{W} \in \mathcal{S}$, then $\mathrm{tr}(\mathbf{W}) = d$ and $0 \le w_{ii} \le 1$.*

Proof. If $\mathbf{W} \in \mathcal{S}$, then $\mathbf{W} = \mathbf{U}\mathbf{U}^T$ with $\mathbf{U}^T\mathbf{U} = \mathbf{I}_d$. Then $\mathrm{tr}(\mathbf{W}) = \mathrm{tr}(\mathbf{U}\mathbf{U}^T) = \mathrm{tr}(\mathbf{U}^T\mathbf{U}) = \mathrm{tr}(\mathbf{I}_d) = d$, and $w_{ii} = \mathbf{e}_i^T\mathbf{W}\mathbf{e}_i = \mathbf{e}_i^T\mathbf{U}\mathbf{U}^T\mathbf{e}_i = \|\mathbf{U}^T\mathbf{e}_i\|^2$, the squared norm of the i-th row of $\mathbf{U}$, which is between 0 and 1. □

By the lemma, the maximum value of $f(\mathbf{W})$, as given by the sum on the right side of (32), occurs when $w_{11} = \cdots = w_{dd} = 1$, and $w_{ii} = 0$ for $i > d$; this is achieved by the matrix

$$\mathbf{W} = \left[\begin{array}{c|c} \mathbf{I}_d & \mathbf{0} \\ \hline \mathbf{0} & \mathbf{0} \end{array}\right] \tag{33}$$

which is contained in $\mathcal{S}$; since $f(\mathbf{W}) = \sum_{i=1}^{d} \lambda_i$, the proof of Ky Fan's Theorem is complete. □

We now prove Theorem 3.1.

Proof of Proposition 3.1. By adjusting the vectors $\boldsymbol{\beta}_i$, we can assume that $\sum_{i=1}^{n} \boldsymbol{\beta}_i = \mathbf{0}$ (replace $\boldsymbol{\beta}_i$ by $\boldsymbol{\beta}_i - \mathbf{a}$, and $\mathbf{c}$ by $\mathbf{c} + \mathbf{U}\mathbf{a}$, where $\mathbf{a} = \sum_{i=1}^{n} \boldsymbol{\beta}_i / n$). The function (24) is convex in $\mathbf{c}$, and taking the derivative of (24) with respect to c_j results in the equation:

$$\begin{aligned} & 2\sum_{i=1}^{n} (c_j + (\mathbf{U}\boldsymbol{\beta}_i)_j - x_{ij}) = 0, \quad 1 \le j \le p \\ \Longrightarrow\ & \mathbf{0} = \sum_{i=1}^{n} (\mathbf{c} + \mathbf{U}\boldsymbol{\beta}_i - \mathbf{x}_i) = n\mathbf{c} - \sum_{i=1}^{n} \mathbf{x}_i, \\ \Longrightarrow\ & \mathbf{c} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i = \boldsymbol{\mu}, \end{aligned} \tag{34}$$

where we have used that $\sum_{i=1}^{n} \boldsymbol{\beta}_i = \mathbf{0}$. Consequently, the optimal $\mathbf{c}$ is $\boldsymbol{\mu}$, the mean.

Let $\widetilde{\mathbf{x}}_i = \mathbf{x}_i - \boldsymbol{\mu}$, the centered observations. To find the optimal $\boldsymbol{\beta}_i$, given $\mathbf{U}$, observe that the objective function (24) may be written as

$$\phi = \sum_{i=1}^{n} \|\widetilde{\mathbf{x}}_i - \mathbf{U}\boldsymbol{\beta}_i\|^2 = \sum_{i=1}^{n} \left(\|\widetilde{\mathbf{x}}_i\|^2 + \|\mathbf{U}\boldsymbol{\beta}_i\|^2 - 2\langle \boldsymbol{\beta}_i, \mathbf{U}^T\widetilde{\mathbf{x}}_i\rangle\right) \tag{35}$$

and taking the gradient with respect to $\boldsymbol{\beta}_i$ yields:

$$\begin{aligned} \mathbf{0} &= 2\mathbf{U}^T\mathbf{U}\boldsymbol{\beta}_i - 2\mathbf{U}^T\widetilde{\mathbf{x}}_i = 2\boldsymbol{\beta}_i - 2\mathbf{U}^T\widetilde{\mathbf{x}}_i \\ \implies \boldsymbol{\beta}_i &= \mathbf{U}^T\widetilde{\mathbf{x}}_i. \end{aligned} \tag{36}$$

Finally, solve for the optimal $\mathbf{U}$. Let $\widetilde{\mathbf{X}} = [\widetilde{\mathbf{x}}_1, \dots, \widetilde{\mathbf{x}}_n]$ denote the matrix of centered data. Note that $n\boldsymbol{\Sigma} = \widetilde{\mathbf{X}}\widetilde{\mathbf{X}}^T$. Rewrite the objective function:

$$\begin{aligned} \phi &= \sum_{i=1}^{n} \|\widetilde{\mathbf{x}}_i - \mathbf{U}\mathbf{U}^T\widetilde{\mathbf{x}}_i\|^2 \\ &= \sum_{i=1}^{n} \|(\mathbf{I}_p - \mathbf{U}\mathbf{U}^T)\widetilde{\mathbf{x}}_i\|^2 \\ &= \|(\mathbf{I}_p - \mathbf{U}\mathbf{U}^T)\widetilde{\mathbf{X}}\|_{\mathrm{F}}^2. \end{aligned} \tag{37}$$

The matrix $\mathbf{P} = \mathbf{I}_p - \mathbf{U}\mathbf{U}^T$ is the orthogonal projection onto the subspace orthogonal to $\mathrm{col}(\mathbf{U})$; consequently, $\mathbf{P}^2 = \mathbf{P}$ and $\mathbf{P}^T = \mathbf{P}$. Continuing:

$$\begin{aligned} \phi &= \|\mathbf{P}\widetilde{\mathbf{X}}\|_{\mathrm{F}}^2 \\ &= \mathrm{tr}(\mathbf{P}\widetilde{\mathbf{X}}\widetilde{\mathbf{X}}^T\mathbf{P}) \\ &= n \cdot \mathrm{tr}(\mathbf{P}\boldsymbol{\Sigma}\mathbf{P}) \\ &= n \cdot \mathrm{tr}(\boldsymbol{\Sigma}\mathbf{P}^2) \\ &= n \cdot \mathrm{tr}(\boldsymbol{\Sigma}\mathbf{P}) \\ &= n \cdot \mathrm{tr}(\boldsymbol{\Sigma}(\mathbf{I}_p - \mathbf{U}\mathbf{U}^T)) \\ &= n \cdot \mathrm{tr}(\boldsymbol{\Sigma}) - n \cdot \mathrm{tr}(\boldsymbol{\Sigma}\mathbf{U}\mathbf{U}^T)) \\ &= n \cdot \mathrm{tr}(\boldsymbol{\Sigma}) - n \cdot \mathrm{tr}(\mathbf{U}^T\boldsymbol{\Sigma}\mathbf{U})) \end{aligned} \tag{38}$$

Consequently, minimizing ϕ over $\mathbf{U}$ is the same as maximizing $\mathrm{tr}(\mathbf{U}^T\boldsymbol{\Sigma}\mathbf{U})$ over $\mathbf{U}$, subject to the constraint $\mathbf{U}^T\mathbf{U} = \mathbf{I}_d$. The result now follows by Ky Fan's Theorem. □

4 Finite sample theory

Let's suppose that we observe n realizations of the random vector X, call them $X_1, \dots, X_n$, and we wish to estimate the PCs of X from these. We consider a standard estimator of $\boldsymbol{\Sigma}$ called the *sample covariance matrix*, defined as

$$\widehat{\boldsymbol{\Sigma}}_n = \frac{1}{n}\sum_{k=1}^{n}(X_k - \widehat{\boldsymbol{\mu}}_n)(X_k - \widehat{\boldsymbol{\mu}}_n)^T. \tag{39}$$

Here, $\widehat{\boldsymbol{\mu}}_n$ denotes the *sample mean*:

$$\widehat{\boldsymbol{\mu}}_n = \frac{1}{n}\sum_{k=1}^{n} X_k. \tag{40}$$

When X is Gaussian, $\widehat{\boldsymbol{\mu}}_n$ and $\widehat{\boldsymbol{\Sigma}}_n$ are the maximum likelihood estimators of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, respectively, and hence are known to have many favorable statistical properties.

From the multivariate central limit theorem, supposing only that X has two finite moments (i.e. that its covariance is well-defined in the first place),

$$\sqrt{n}\left(\frac{1}{n}\sum_{k=1}^{n}(X_k-\boldsymbol{\mu})(X_k-\boldsymbol{\mu})^T-\boldsymbol{\Sigma}\right)\longrightarrow^d \mathcal{N}(\mathbf{0},\mathbf{A}), \tag{41}$$

for some positive definite $\mathbf{A}\in\mathbb{R}^{p^2\times p^2}$, where the convergence is in distribution. Since the same can be said of $\widehat{\boldsymbol{\mu}}_n$ converging to $\boldsymbol{\mu}$, it follows (check this!) that

$$\sqrt{n}\left(\widehat{\boldsymbol{\Sigma}}_n-\boldsymbol{\Sigma}\right)\longrightarrow^d \mathcal{N}(\mathbf{0},\mathbf{A}). \tag{42}$$

If X has finite fourth moments, then the convergence is almost sure, by the strong law of large numbers.

Given that the sample covariance $\widehat{\boldsymbol{\Sigma}}_n$ is "close" to the population covariance $\boldsymbol{\Sigma}$, what can be said about the proximity of their eigenvectors? This is answered by the celebrated Davis-Kahan $\sin\Theta$ Theorem, which we state and prove in the next section.

5 The Davis-Kahan Theorem

We first define the *principal angles* between subspaces, as follows.

Let $\mathbf{V}=[\mathbf{v}_1,\dots,\mathbf{v}_d]$ and $\widehat{\mathbf{V}}=[\widehat{\mathbf{v}}_1,\dots,\widehat{\mathbf{v}}_d]$ be two p-by-d matrices, each with orthonormal columns. Then the principal angles between the subspaces $\text{span}\{\mathbf{v}_1,\dots,\mathbf{v}_d\}$ and $\text{span}\{\widehat{\mathbf{v}}_1,\dots,\widehat{\mathbf{v}}_d\}$ are the numbers $\theta_k=\cos^{-1}(\eta_k)$, $1\le k\le d$, where $\eta_1,\dots,\eta_d$ are the singular values of $\widehat{\mathbf{V}}^T\mathbf{V}$. Then $\Theta(\widehat{\mathbf{V}},\mathbf{V})=\text{diag}(\theta_1,\dots,\theta_d)$, and $\sin\Theta(\widehat{\mathbf{V}},\mathbf{V})$ is defined entry-wise. Note that

$$\|\sin\Theta(\widehat{\mathbf{V}},\mathbf{V})\|_{\text{F}}^2=d-\|\widehat{\mathbf{V}}^T\mathbf{V}\|_{\text{F}}^2; \tag{43}$$

indeed, since $\eta_1,\dots,\eta_d$ are the singular values of $\widehat{\mathbf{V}}^T\mathbf{V}$,

$$\|\widehat{\mathbf{V}}^T\mathbf{V}\|_{\text{F}}^2=\sum_{k=1}^{d}\eta_k^2=\sum_{k=1}^{d}\cos^2(\theta_k)=\sum_{k=1}^{d}(1-\sin^2(\theta_k))=d-\sum_{k=1}^{d}\sin^2(\theta_k)=d-\|\sin\Theta(\widehat{\mathbf{V}},\mathbf{V})\|_{\text{F}}^2. \tag{44}$$

The interpretation of the principal angles may be found from the following result, which is analogous to Rayleigh-Ritz:

Theorem 5.1. *Let $\mathbf{A}$ be any real m-by-n matrix, $m\le n$, with left singular vectors $\mathbf{u}_1,\dots,\mathbf{u}_m$, right singular vectors $\mathbf{v}_1,\dots,\mathbf{v}_m$, and singular values $\sigma_1\ge\cdots\ge\sigma_m\ge 0$. Then*

$$\begin{aligned}\sigma_1&=\max_{\substack{\mathbf{u}\in\mathbb{R}^m:\\ \|\mathbf{u}\|=1}}\max_{\substack{\mathbf{v}\in\mathbb{R}^n:\\ \|\mathbf{v}\|=1}}\mathbf{u}^T\mathbf{A}\mathbf{v}\\ &=\mathbf{u}_1^T\mathbf{A}\mathbf{v}_1,\end{aligned} \tag{45}$$

and for $k\ge 2$,

$$\begin{aligned}\sigma_k&=\max_{\substack{\mathbf{u}\in\mathbb{R}^m:\|\mathbf{u}\|=1,\\ \mathbf{u}\perp\mathbf{u}_1,\dots,\mathbf{u}_{k-1}}}\max_{\substack{\mathbf{v}\in\mathbb{R}^n:\|\mathbf{v}\|=1,\\ \mathbf{v}\perp\mathbf{v}_1,\dots,\mathbf{v}_{k-1}}}\mathbf{u}^T\mathbf{A}\mathbf{v}\\ &=\mathbf{u}_k^T\mathbf{A}\mathbf{v}_k.\end{aligned} \tag{46}$$

Proof. Exercise. □

With this theorem, we see that the principal angle θ_1 is the smallest angle between any unit vector in $\text{col}(\mathbf{V})$ and any unit vector in $\text{col}(\widehat{\mathbf{V}})$; if $\mathbf{u}_1$ and $\widehat{\mathbf{u}}_1$ are the corresponding vectors with angle θ_1, then the principal angle θ_2 is the smallest angle between any unit vector in $\text{col}(\mathbf{V})\cap\text{span}\{\mathbf{u}_1\}^\perp$ and any unit vector $\text{col}(\widehat{\mathbf{V}})\cap\text{span}\{\widehat{\mathbf{u}}_1\}^\perp$; if $\mathbf{u}_2$ and $\widehat{\mathbf{u}}_2$ are the corresponding vectors with angle θ_2, then the principal angle θ_3 is the smallest angle between any unit vector in $\text{col}(\mathbf{V})\cap\text{span}\{\mathbf{u}_1,\mathbf{u}_2\}^\perp$ and any unit vector $\text{col}(\widehat{\mathbf{V}})\cap\text{span}\{\widehat{\mathbf{u}}_1,\widehat{\mathbf{u}}_2\}^\perp$; etc. Of course, if $d=1$, then $\sin\Theta(\widehat{\mathbf{v}},\mathbf{v})=\sqrt{1-(\widehat{\mathbf{v}}^T\mathbf{v})^2}$, the usual definition of the sine of the angle between unit vectors.

With this definition, we now state the theorem:

Theorem 5.2 (Davis-Kahan $\sin\Theta$ Theorem for Statisticians). *Suppose* $\mathbf{A}$ *and* $\widehat{\mathbf{A}}$ *are two real, symmetric p-by-p matrices. Denote the eigenvalues of* $\mathbf{A}$ *by* $\lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_p$, *and the eigenvalues of* $\widehat{\mathbf{A}}$ *by* $\widehat{\lambda}_1 \ge \widehat{\lambda}_2 \ge \cdots \ge \widehat{\lambda}_p$; *and corresponding orthonormal eigenvectors by* $\mathbf{v}_1, \dots, \mathbf{v}_p$ *and* $\widehat{\mathbf{v}}_1, \dots, \widehat{\mathbf{v}}_p$, *respectively. For any* $1 \le r \le s \le p$, *let* $d = s - r + 1$, *and define the p-by-d matrices*

$$\mathbf{V} = [\mathbf{v}_r, \dots, \mathbf{v}_s] \tag{47}$$

and

$$\widehat{\mathbf{V}} = [\widehat{\mathbf{v}}_r, \dots, \widehat{\mathbf{v}}_s]. \tag{48}$$

Then:

$$\|\sin\Theta(\widehat{\mathbf{V}}, \mathbf{V})\|_{\mathrm{F}} \le 2\frac{\min\{\sqrt{d}\|\widehat{\mathbf{A}} - \mathbf{A}\|_{\mathrm{op}}, \|\widehat{\mathbf{A}} - \mathbf{A}\|_{\mathrm{F}}\}}{\min\{\lambda_{r-1} - \lambda_r, \lambda_s - \lambda_{s+1}\}}. \tag{49}$$

where $\lambda_0 = \infty$ *and* $\lambda_{p+1} = -\infty$.

This version of the Davis-Kahan Theorem, along with the proof we will give, come from the paper [6].

It is natural to estimate the principal components of X via the principal components of the sample covariance $\widehat{\boldsymbol{\Sigma}}_n$. Davis-Kahan ensures that the eigenvectors of $\widehat{\boldsymbol{\Sigma}}_n$ converge to the PCs of X.

5.1 General results used in the proof of Davis-Kahan

We start with several results that are of general interest, and which are used in the proof of Davis-Kahan. The first is the Wielandt-Hoffman Theorem:

Theorem 5.3 (Wielandt-Hoffman). *Let* $\mathbf{A}$ *and* $\mathbf{B}$ *be symmetric n-by-n matrices with eigenvalues* $\lambda_1 \ge \cdots \ge \lambda_n$ *and* $\nu_1 \ge \cdots \ge \nu_n$, *respectively. Then*

$$\sum_{j=1}^{n} (\lambda_j - \nu_j)^2 \le \|\mathbf{A} - \mathbf{B}\|_F^2. \tag{50}$$

The proof of Wielandt-Hoffman requires the Birkhoff-Von Neumann Theorem, which we will state here but not prove. A square matrix $\mathbf{Q}$ is said to be *bistochastic* if $\mathbf{Q}\mathbf{1} = \mathbf{Q}^T\mathbf{1} = \mathbf{1}$ and all entries of $\mathbf{Q}$ are non-negative. Also, recall that an n-by-n permutation matrix $\mathbf{P} = \mathbf{P}_\pi$ corresponding to a permutation π of $\{1, \dots, n\}$ has entries $\mathbf{P}_{\pi(j)j} = 1$ for all j and all other entries are zero. With this terminology, we have:

Theorem 5.4 (Birkhoff-Von Neumann). *Let* $\mathcal{B}$ *denote the set of n-by-n bistochastic matrices. Then* $\mathcal{B}$ *is convex, and the extrema of* $\mathcal{B}$ *are the permutation matrices.*

Proof of Wielandt-Hoffman. Write $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^T$ and $\mathbf{B} = \mathbf{V}\mathbf{E}\mathbf{V}^T$, where $\mathbf{D} = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$ and $\mathbf{E} = \mathrm{diag}(\nu_1, \dots, \nu_n)$. Then defining $\mathbf{Q} = \mathbf{V}^T\mathbf{U}$, we have:

$$\|\mathbf{A} - \mathbf{B}\|_{\mathrm{F}}^2 = \|\mathbf{U}\mathbf{D}\mathbf{U}^T - \mathbf{V}\mathbf{E}\mathbf{V}^T\|_{\mathrm{F}}^2 = \|\mathbf{V}^T\mathbf{U}\mathbf{D} - \mathbf{E}\mathbf{V}^T\mathbf{U}\|_{\mathrm{F}}^2 = \|\mathbf{Q}\mathbf{D} - \mathbf{E}\mathbf{Q}\|_{\mathrm{F}}^2 = \sum_{i,j} q_{ij}^2(\lambda_j - \nu_i)^2. \tag{51}$$

The matrix (q_{ij}^2) is bistochastic, since $\mathbf{Q}$ is an orthogonal matrix. If we let $\mathcal{B}$ be the set of bistochastic matrices, we can therefore write

$$\|\mathbf{A} - \mathbf{B}\|_{\mathrm{F}}^2 \ge \min_{\mathbf{P} \in \mathcal{B}} \Phi(\mathbf{P}), \tag{52}$$

where $\Phi(\mathbf{P})$ is the linear functional $\Phi(\mathbf{P}) = \sum_{i,j} p_{ij}(\lambda_j - \nu_i)^2$. We will show that $\min_{\mathbf{P} \in \mathcal{B}} \Phi(\mathbf{P}) = \sum_{j=1}^{n} (\lambda_j - \nu_j)^2 = \Phi(\mathbf{I})$, i.e. the minimum of Φ is achieved at the identity matrix $\mathbf{I}$.

First, the set $\mathcal{B}$ is convex, and so the minimum of the linear functional Φ must be achieved at its extreme points. The Birkhoff-Von Neumann Theorem states that the extrema of $\mathcal{B}$ are the permutation matrices. Consequently, it is enough to show that $\Phi(\mathbf{I}) \le \Phi(\mathbf{P})$ for any permutation matrix $\mathbf{P}$.

Take any permutation π of $1, \dots, n$. If π is not the identity, then let j be the first index such that $\pi(j) \neq j$. Without loss of generality, assume $j = 1$. Let $k = \pi(1)$; so $k > 1$; and without loss of generality assume $\pi(2) = 1$. Then define a new permutation π' so that $\pi'(1) = 1$, $\pi'(2) = k$, but otherwise π' agrees with π. Then π' has at least one more fixed point than π, and so is closer to the identity; consequently, if $\mathbf{P}_\pi$ and $\mathbf{P}_{\pi'}$ are corresponding permutation matrices, we would like to show that $\Phi(\mathbf{P}_{\pi'}) \le \Phi(\mathbf{P}_\pi)$.

We have

$$\Phi(\mathbf{P}_\pi) = \sum_{j=1}^{n} (\lambda_j - \nu_{\pi(j)})^2 = (\lambda_1 - \nu_k)^2 + (\lambda_2 - \nu_1)^2 + \sum_{j=3}^{n} (\lambda_j - \nu_{\pi(j)})^2 \tag{53}$$

and

$$\Phi(\mathbf{P}_{\pi'}) = \sum_{j=1}^{n} (\lambda_j - \nu_{\pi'(j)})^2 = (\lambda_1 - \nu_1)^2 + (\lambda_2 - \nu_k)^2 + \sum_{j=3}^{n} (\lambda_j - \nu_{\pi'(j)})^2. \tag{54}$$

So we want to show that

$$(\lambda_1 - \nu_1)^2 + (\lambda_2 - \nu_k)^2 \le (\lambda_1 - \nu_k)^2 + (\lambda_2 - \nu_1)^2. \tag{55}$$

To that end, we may write

$$(\lambda_1 - \nu_k)^2 = (\lambda_1 - \nu_1 + \nu_1 - \nu_k)^2 = (\lambda_1 - \nu_1)^2 + (\nu_1 - \nu_k)^2 + 2(\lambda_1 - \nu_1)(\nu_1 - \nu_k) \tag{56}$$

and

$$(\lambda_2 - \nu_1)^2 = (\lambda_2 - \nu_k + \nu_k - \nu_1)^2 = (\lambda_2 - \nu_k)^2 + (\nu_k - \nu_1)^2 + 2(\lambda_2 - \nu_k)(\nu_k - \nu_1) \tag{57}$$

and adding these we get:

$$\begin{aligned}
&(\lambda_1 - \nu_k)^2 + (\lambda_2 - \nu_1)^2 \\
&= (\lambda_1 - \nu_1)^2 + (\nu_1 - \nu_k)^2 + 2(\lambda_1 - \nu_1)(\nu_1 - \nu_k) + (\lambda_2 - \nu_k)^2 + (\nu_k - \nu_1)^2 + 2(\lambda_2 - \nu_k)(\nu_k - \nu_1) \\
&= (\lambda_1 - \nu_1)^2 + (\lambda_2 - \nu_k)^2 + (\nu_1 - \nu_k)^2 + 2(\lambda_1 - \nu_1)(\nu_1 - \nu_k) + (\nu_k - \nu_1)^2 + 2(\lambda_2 - \nu_k)(\nu_k - \nu_1) \\
&= (\lambda_1 - \nu_1)^2 + (\lambda_2 - \nu_k)^2 + (\nu_1 - \nu_k)(\nu_1 - \nu_k + 2\lambda_1 - 2\nu_1 - \nu_k + \nu_1 - 2\lambda_2 + 2\nu_k) \\
&= (\lambda_1 - \nu_1)^2 + (\lambda_2 - \nu_k)^2 + 2(\nu_1 - \nu_k)(\lambda_1 - \lambda_2) \\
&\ge (\lambda_1 - \nu_1)^2 + (\lambda_2 - \nu_k)^2,
\end{aligned} \tag{58}$$

since $\nu_1 \ge \nu_k$ and $\lambda_1 \ge \lambda_2$. This completes the proof. □

Next is Weyl's Theorem, which is analogous to Wielandt-Hoffman but with the infinity norm replacing the 2 norm:

Theorem 5.5 (Weyl)**.** *Let $\mathbf{A}$ and $\mathbf{B}$ be symmetric n-by-n matrices with eigenvalues $\lambda_1 \ge \dots \ge \lambda_n$ and $\nu_1 \ge \dots \ge \nu_n$, respectively. Then*

$$\max_{1 \le j \le n} |\lambda_j - \nu_j| \le \|\mathbf{A} - \mathbf{B}\|_{\mathrm{op}}. \tag{59}$$

Proof of Weyl's Theorem. Let $\mathbf{E} = \mathbf{B} - \mathbf{A}$, and let $\epsilon_1 \ge \dots \ge \epsilon_n$ be the eigenvalues of $\mathbf{E}$. By Courant-Hilbert, there is a k-dimensional subspace $\mathcal{H}_k \le \mathbb{R}^n$ (namely, the span of the top k eigenvectors of $\mathbf{B}$) such that

$$\nu_k = \min_{\substack{\mathbf{v} \in \mathcal{H}_k: \\ \|\mathbf{v}\|=1}} \mathbf{v}^T \mathbf{B} \mathbf{v}. \tag{60}$$

Also by Courant-Hilbert,

$$\lambda_k = \max_{\substack{\mathcal{H} \le \mathbb{R}^n: \\ \dim \mathcal{H} = k}} \min_{\substack{\mathbf{v} \in \mathcal{H}: \\ \|\mathbf{v}\|=1}} \mathbf{v}^T \mathbf{A} \mathbf{v} \ge \min_{\substack{\mathbf{v} \in \mathcal{H}_k: \\ \|\mathbf{v}\|=1}} \mathbf{v}^T \mathbf{A} \mathbf{v} = \overline{\mathbf{v}}^T \mathbf{A} \overline{\mathbf{v}}, \tag{61}$$

for some vector $\overline{\mathbf{v}} \in \mathcal{H}_k$. Consequently,

$$\nu_k = \min_{\substack{\mathbf{v}\in\mathcal{H}_k:\\ \|\mathbf{v}\|=1}} \mathbf{v}^T\mathbf{B}\mathbf{v} \le \overline{\mathbf{v}}^T\mathbf{B}\overline{\mathbf{v}} = \overline{\mathbf{v}}^T(\mathbf{A}+\mathbf{E})\overline{\mathbf{v}} = \overline{\mathbf{v}}^T\mathbf{A}\overline{\mathbf{v}} + \overline{\mathbf{v}}^T\mathbf{E}\overline{\mathbf{v}} \le \lambda_k + \overline{\mathbf{v}}^T\mathbf{E}\overline{\mathbf{v}} \le \lambda_k + \epsilon_1, \tag{62}$$

where we have used that ϵ_1, the largest eigenvalue of $\mathbf{E}$, is the maximum of $\mathbf{v}^T\mathbf{E}\mathbf{v}$ over all unit vectors $\mathbf{v}$.

Switching the roles of $\mathbf{A}$ and $\mathbf{B}$ and replacing $\mathbf{E}$ with $-\mathbf{E}$, this also shows:

$$\lambda_k \le \nu_k - \epsilon_n. \tag{63}$$

Consequently, we have shown $\lambda_k + \epsilon_n \le \nu_k \le \lambda_k + \epsilon_1$, or equivalently,

$$\epsilon_n \le \nu_k - \lambda_k \le \epsilon_1. \tag{64}$$

Since $\|\mathbf{A}-\mathbf{B}\|_{\mathrm{op}} = \|\mathbf{E}\|_{\mathrm{op}} = \max\{|\epsilon_1|, |\epsilon_n|\}$, this proves

$$\max_k |\nu_k - \lambda_k| \le \max\{|\epsilon_1|, |\epsilon_n|\} = \|\mathbf{A}-\mathbf{B}\|_{\mathrm{op}}, \tag{65}$$

completing the proof. □

5.2 Additional lemmas used in the proof of Davis-Kahan

Lemma 5.6. *Suppose $\mathbf{A} \in \mathbb{R}^{p\times n}$, and $\mathbf{U} \in \mathbb{R}^{p\times d}$ and $\mathbf{W} \in \mathbb{R}^{n\times r}$ have orthonormal rows. Then $\|\mathbf{U}^T\mathbf{A}\mathbf{W}\|_{\mathrm{F}} = \|\mathbf{A}\|_{\mathrm{F}}$.*

Proof. Using $\|\mathbf{B}\|_{\mathrm{F}}^2 = \mathrm{tr}(\mathbf{B}\mathbf{B}^T)$ the cyclic property of traces, and $\mathbf{W}\mathbf{W}^T = \mathbf{I}_n$ and $\mathbf{U}\mathbf{U}^T = \mathbf{I}_p$,

$$\|\mathbf{U}^T\mathbf{A}\mathbf{W}\|_{\mathrm{F}}^2 = \mathrm{tr}(\mathbf{U}^T\mathbf{A}\mathbf{W}\mathbf{W}^T\mathbf{A}^T\mathbf{U}) = \mathrm{tr}(\mathbf{A}\mathbf{W}\mathbf{W}^T\mathbf{A}^T\mathbf{U}\mathbf{U}^T) = \mathrm{tr}(\mathbf{A}\mathbf{A}^T) = \|\mathbf{A}\|_{\mathrm{F}}^2, \tag{66}$$

which is the desired result. □

Lemma 5.7. *Suppose $\mathbf{A} \in \mathbb{R}^{p\times n}$, and $\mathbf{U} \in \mathbb{R}^{p\times d}$ and $\mathbf{V} \in \mathbb{R}^{n\times r}$ have orthonormal columns. Then $\|\mathbf{U}^T\mathbf{A}\mathbf{W}\|_{\mathrm{F}} \le \|\mathbf{A}\|_{\mathrm{F}}$.*

Proof. Take $\widetilde{\mathbf{U}}$ and $\widetilde{\mathbf{W}}$ such that $[\mathbf{U}\ \ \widetilde{\mathbf{U}}]$ and $[\mathbf{W}\ \ \widetilde{\mathbf{W}}]$ are orthogonal. Then

$$\begin{aligned}
\|\mathbf{A}\|_{\mathrm{F}}^2 &= \left\| \begin{bmatrix} \mathbf{U}^T \\ \widetilde{\mathbf{U}}^T \end{bmatrix} \mathbf{A} \begin{bmatrix}\mathbf{W} & \widetilde{\mathbf{W}}\end{bmatrix} \right\|_{\mathrm{F}}^2 \\
&= \left\| \begin{bmatrix} \mathbf{U}^T\mathbf{A}\mathbf{W} & \mathbf{U}^T\mathbf{A}\widetilde{\mathbf{W}} \\ \widetilde{\mathbf{U}}^T\mathbf{A}\mathbf{W} & \widetilde{\mathbf{U}}^T\mathbf{A}\widetilde{\mathbf{W}} \end{bmatrix} \right\|_{\mathrm{F}}^2 \\
&= \|\mathbf{U}^T\mathbf{A}\mathbf{W}\|_{\mathrm{F}^2}^2 + \|\mathbf{U}^T\mathbf{A}\widetilde{\mathbf{W}}\|_{\mathrm{F}^2}^2 \|\widetilde{\mathbf{U}}^T\mathbf{A}\mathbf{W}\|_{\mathrm{F}^2}^2 + \|\widetilde{\mathbf{U}}^T\mathbf{A}\widetilde{\mathbf{W}}\|_{\mathrm{F}^2}^2 \\
&\ge \left\|\mathbf{U}^T\mathbf{A}\mathbf{W}\right\|_{\mathrm{F}}^2,
\end{aligned} \tag{67}$$

completing the proof. □

Lemma 5.8. *Let $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ be three matrices with dimensions so that the product $\mathbf{ABC}$ is well-defined. Let $\otimes$ denote the Kronecker product, and let* vec *denote the vectorization operator that stacks a matrix's columns. Then*

$$\mathrm{vec}(\mathbf{ABC}) = (\mathbf{C}^T \otimes \mathbf{A})\,\mathrm{vec}(\mathbf{B}). \tag{68}$$

Proof. Suppose $\mathbf{A}$ is of size p-by-n. By definition of matrix multiplication,

$$(\mathbf{ABC})_{ik} = \sum_{j,l} c_{jk} a_{il} b_{lj}. \tag{69}$$

By definition of the vectorization operation,

$$b_{lj} = \mathrm{vec}(\mathbf{B})_{n(j-1)+l}; \tag{70}$$

and

$$(\mathbf{ABC})_{ik} = \mathrm{vec}(\mathbf{ABC})_{p(k-1)+i}. \tag{71}$$

By definition of the Kronecker product,

$$c_{jk} a_{il} = (\mathbf{C}^T \otimes \mathbf{A})_{p(k-1)+i, n(j-1)+l}. \tag{72}$$

Consequently,

$$\begin{aligned}
\mathrm{vec}(\mathbf{ABC})_{p(k-1)+i} &= (\mathbf{ABC})_{ik} \\
&= \sum_{j,l} c_{jk} a_{il} b_{lj} \\
&= \sum_{j,l} (\mathbf{C}^T \otimes \mathbf{A})_{p(k-1)+i, n(j-1)+l} \,\mathrm{vec}(\mathbf{B})_{n(j-1)+l} \\
&= ((\mathbf{C}^T \otimes \mathbf{A}) \,\mathrm{vec}(\mathbf{B}))_{p(k-1)+i},
\end{aligned} \tag{73}$$

completing the proof.

□

5.3 Proof of Davis-Kahan

With the background results in hand, we now turn to the proof of Davis-Kahan itself. First define some auxiliary notation. Define the diagonal matrices of eigenvalues:

$$\mathbf{\Lambda} = \mathrm{diag}(\lambda_r, \dots, \lambda_s) \tag{74}$$

and

$$\mathbf{\Lambda}_1 = \mathrm{diag}(\lambda_1, \dots, \lambda_{r-1}, \lambda_s, \dots, \lambda_p). \tag{75}$$

Also define

$$\mathbf{V}_1 = [\mathbf{v}_1, \dots, \mathbf{v}_{r-1}, \mathbf{v}_{s+1}, \dots, \mathbf{v}_p], \tag{76}$$

and recalling that $\mathbf{V} = [\mathbf{v}_r, , \dots, \mathbf{v}_s]$, we note that

$$\mathbf{I}_p = \mathbf{V}\mathbf{V}^T + \mathbf{V}_1 \mathbf{V}_1^T. \tag{77}$$

First, we will bound $\|\widehat{\mathbf{V}}\mathbf{\Lambda} - \mathbf{A}\widehat{\mathbf{V}}\|_{\mathrm{F}}$, using Weyl's Theorem and Wielandt-Hoffman:

Proposition 5.9. *With the notation as in the statement of Davis-Kahan,*

$$\|\widehat{\mathbf{V}}\mathbf{\Lambda} - \mathbf{A}\widehat{\mathbf{V}}\|_{\mathrm{F}} \le 2\min\{\sqrt{d}\|\widehat{\mathbf{A}} - \mathbf{A}\|_{\mathrm{op}}, \|\widehat{\mathbf{A}} - \mathbf{A}\|_{\mathrm{F}}\}. \tag{78}$$

Proof of Proposition 5.9. By the definition of eigenvalues and eigenvectors, $\widehat{\mathbf{A}}\widehat{\mathbf{V}} = \widehat{\mathbf{V}}\widehat{\mathbf{\Lambda}}$, and so

$$\mathbf{0} = \widehat{\mathbf{A}}\widehat{\mathbf{V}} - \widehat{\mathbf{V}}\widehat{\mathbf{\Lambda}} = \mathbf{A}\widehat{\mathbf{V}} - \widehat{\mathbf{V}}\mathbf{\Lambda} + (\widehat{\mathbf{A}} - \mathbf{A})\widehat{\mathbf{V}} - \widehat{\mathbf{V}}(\widehat{\mathbf{\Lambda}} - \mathbf{\Lambda}), \tag{79}$$

and so

$$\mathbf{A}\widehat{\mathbf{V}} - \widehat{\mathbf{V}}\mathbf{\Lambda} = \widehat{\mathbf{V}}(\widehat{\mathbf{\Lambda}} - \mathbf{\Lambda}) - (\widehat{\mathbf{A}} - \mathbf{A})\widehat{\mathbf{V}}. \tag{80}$$

Consequently,

$$\begin{aligned}
\|\mathbf{A}\widehat{\mathbf{V}} - \widehat{\mathbf{V}}\mathbf{\Lambda}\|_{\mathrm{F}} &\le \|\widehat{\mathbf{V}}(\widehat{\mathbf{\Lambda}} - \mathbf{\Lambda})\|_{\mathrm{F}} + \|(\widehat{\mathbf{A}} - \mathbf{A})\widehat{\mathbf{V}}\|_{\mathrm{F}} \\
&= \|\widehat{\mathbf{\Lambda}} - \mathbf{\Lambda}\|_{\mathrm{F}} + \|(\widehat{\mathbf{A}} - \mathbf{A})\widehat{\mathbf{V}}\|_{\mathrm{F}} \\
&\le \|\widehat{\mathbf{\Lambda}} - \mathbf{\Lambda}\|_{\mathrm{F}} + \|\widehat{\mathbf{A}} - \mathbf{A}\|_{\mathrm{F}} \\
&\le 2\|\widehat{\mathbf{A}} - \mathbf{A}\|_{\mathrm{F}},
\end{aligned} \tag{81}$$

where in the last inequality we have used the Wielandt-Hoffman Theorem, and in the second-to-last inequality we have used Lemma 5.7.

For the other part of the bound, observe first that

$$\|(\widehat{\mathbf{A}} - \mathbf{A})\widehat{\mathbf{V}}\|_{\mathrm{F}}^2 = \sum_{j=1}^{d} \|(\widehat{\mathbf{A}} - \mathbf{A})\widehat{\mathbf{v}}_j\|_{\mathrm{F}}^2 \le \sum_{j=1}^{d} \|\widehat{\mathbf{A}} - \mathbf{A}\|_{\mathrm{op}}^2 = d \cdot \|\widehat{\mathbf{A}} - \mathbf{A}\|_{\mathrm{op}}^2, \tag{82}$$

and furthermore by Weyl's Theorem

$$\|\widehat{\mathbf{\Lambda}} - \mathbf{\Lambda}\|_{\mathrm{F}} \le \sqrt{d} \max_j |\widehat{\lambda}_j - \lambda_j| \le \sqrt{d}\|\widehat{\mathbf{A}} - \mathbf{A}\|_{\mathrm{op}}. \tag{83}$$

Consequently,

$$\begin{aligned} \|\mathbf{A}\widehat{\mathbf{V}} - \widehat{\mathbf{V}}\mathbf{\Lambda}\|_{\mathrm{F}} &\le \|\widehat{\mathbf{V}}(\widehat{\mathbf{\Lambda}} - \mathbf{\Lambda})\|_{\mathrm{F}} + \|(\widehat{\mathbf{A}} - \mathbf{A})\widehat{\mathbf{V}}\|_{\mathrm{F}} \\ &= \|\widehat{\mathbf{\Lambda}} - \mathbf{\Lambda}\|_{\mathrm{F}} + \|(\widehat{\mathbf{A}} - \mathbf{A})\widehat{\mathbf{V}}\|_{\mathrm{F}} \\ &\le 2\sqrt{d}\|\widehat{\mathbf{A}} - \mathbf{A}\|_{\mathrm{op}}, \end{aligned} \tag{84}$$

completing the proof.

□

We next state another bound

Proposition 5.10. *With the notation as in the statement of Davis-Kahan, we have the bound:*

$$\|\mathbf{V}_1^T\widehat{\mathbf{V}}\mathbf{\Lambda} - \mathbf{\Lambda}_1\mathbf{V}_1^T\widehat{\mathbf{V}}\|_{\mathrm{F}} \le \|\widehat{\mathbf{V}}\mathbf{\Lambda} - \mathbf{A}\widehat{\mathbf{V}}\|_{\mathrm{F}}. \tag{85}$$

Proof of Proposition 5.10. Recall the identity

$$\mathbf{I}_p = \mathbf{V}\mathbf{V}^T + \mathbf{V}_1\mathbf{V}_1^T. \tag{86}$$

Using this identity and $\mathbf{V}^T\mathbf{A} = \mathbf{\Lambda}\mathbf{V}^T$ and $\mathbf{V}_1^T\mathbf{A} = \mathbf{\Lambda}_1\mathbf{V}_1^T$, we find:

$$\begin{aligned} \widehat{\mathbf{V}}\mathbf{\Lambda} - \mathbf{A}\widehat{\mathbf{V}} &= (\mathbf{V}\mathbf{V}^T + \mathbf{V}_1\mathbf{V}_1^T)\widehat{\mathbf{V}}\mathbf{\Lambda} - (\mathbf{V}\mathbf{V}^T + \mathbf{V}_1\mathbf{V}_1^T)\mathbf{A}\widehat{\mathbf{V}} \\ &= \mathbf{V}\mathbf{V}^T\widehat{\mathbf{V}}\mathbf{\Lambda} + \mathbf{V}_1\mathbf{V}_1^T\widehat{\mathbf{V}}\mathbf{\Lambda} - \mathbf{V}\mathbf{V}^T\mathbf{A}\widehat{\mathbf{V}} - \mathbf{V}_1\mathbf{V}_1^T\mathbf{A}\widehat{\mathbf{V}} \\ &= \mathbf{V}\mathbf{V}^T\widehat{\mathbf{V}}\mathbf{\Lambda} + \mathbf{V}_1\mathbf{V}_1^T\widehat{\mathbf{V}}\mathbf{\Lambda} - \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T\widehat{\mathbf{V}} - \mathbf{V}_1\mathbf{\Lambda}_1\mathbf{V}_1^T\widehat{\mathbf{V}} \\ &= \mathbf{V}_1\mathbf{C} + \mathbf{V}\mathbf{D}, \end{aligned} \tag{87}$$

where for brevity we have set $\mathbf{C} = \mathbf{V}_1^T\widehat{\mathbf{V}}\mathbf{\Lambda} - \mathbf{\Lambda}_1\mathbf{V}_1^T\widehat{\mathbf{V}}$ and $\mathbf{D} = \mathbf{V}^T\widehat{\mathbf{V}}\mathbf{\Lambda} - \mathbf{\Lambda}\mathbf{V}^T\widehat{\mathbf{V}}$. Then, since $\mathbf{V}^T\mathbf{V}_1 = \mathbf{0}$,

$$\langle \mathbf{V}_1\mathbf{C}, \mathbf{V}\mathbf{D}\rangle = \mathrm{tr}[\mathbf{D}^T\mathbf{V}^T\mathbf{V}_1\mathbf{C}] = 0. \tag{88}$$

Consequently, since $\mathbf{V}_1^T\mathbf{V}_1 = \mathbf{I}_{p-d}$,

$$\begin{aligned} \|\widehat{\mathbf{V}}\mathbf{\Lambda} - \mathbf{A}\widehat{\mathbf{V}}\|_{\mathrm{F}}^2 &= \|\mathbf{V}_1\mathbf{C} + \mathbf{V}\mathbf{D}\|_{\mathrm{F}}^2 \\ &= \|\mathbf{V}_1\mathbf{C}\|_{\mathrm{F}}^2 + \|\mathbf{V}\mathbf{D}\|_{\mathrm{F}}^2 \\ &\ge \|\mathbf{V}_1\mathbf{C}\|_{\mathrm{F}}^2 \\ &= \|\mathbf{C}\|_{\mathrm{F}}^2 \\ &= \|\mathbf{V}_1^T\widehat{\mathbf{V}}\mathbf{\Lambda} - \mathbf{\Lambda}_1\mathbf{V}_1^T\widehat{\mathbf{V}}\|_{\mathrm{F}}^2. \end{aligned} \tag{89}$$

□

Next, we have the following algebraic identity:

Lemma 5.11. *We have the identity*

$$\|\sin\Theta(\widehat{\mathbf{V}}, \mathbf{V})\|_{\mathrm{F}} = \|\mathrm{vec}(\mathbf{V}_1^T\widehat{\mathbf{V}})\|. \tag{90}$$

Proof of Lemma 5.11. We have:

$$\begin{aligned}\|\operatorname{vec}(\mathbf{V}_1^T\widehat{\mathbf{V}})\|^2 &= \|\mathbf{V}_1^T\widehat{\mathbf{V}}\|_{\mathrm{F}}^2 \\ &= \operatorname{tr}(\widehat{\mathbf{V}}^T\mathbf{V}_1\mathbf{V}_1^T\widehat{\mathbf{V}}) \\ &= \operatorname{tr}(\mathbf{V}_1\mathbf{V}_1^T\widehat{\mathbf{V}}\widehat{\mathbf{V}}^T) \\ &= \operatorname{tr}((\mathbf{I}_p - \mathbf{V}\mathbf{V}^T)\widehat{\mathbf{V}}\widehat{\mathbf{V}}^T) \\ &= d - \operatorname{tr}(\mathbf{V}\mathbf{V}^T\widehat{\mathbf{V}}\widehat{\mathbf{V}}^T) \\ &= d - \|\mathbf{V}^T\widehat{\mathbf{V}}\|_{\mathrm{F}}^2 \\ &= \|\sin\Theta(\widehat{\mathbf{V}},\mathbf{V})\|_{\mathrm{F}}^2, \end{aligned} \tag{91}$$

as claimed.

□

Note that with Proposition 5.9, Proposition 5.10, and Lemma 5.11, Davis-Kahan is reduced to proving

$$\|\operatorname{vec}(\mathbf{V}_1^T\widehat{\mathbf{V}})\| \le \frac{\|\mathbf{V}_1^T\widehat{\mathbf{V}}\boldsymbol{\Lambda} - \boldsymbol{\Lambda}_1\mathbf{V}_1^T\widehat{\mathbf{V}}\|_{\mathrm{F}}}{\min\{\lambda_{r-1}-\lambda_r, \lambda_s - \lambda_{s+1}\}}, \tag{92}$$

or equivalently,

$$\|\mathbf{V}_1^T\widehat{\mathbf{V}}\boldsymbol{\Lambda} - \boldsymbol{\Lambda}_1\mathbf{V}_1^T\widehat{\mathbf{V}}\|_{\mathrm{F}} \ge \min\{\lambda_{r-1}-\lambda_r, \lambda_s - \lambda_{s+1}\} \cdot \|\operatorname{vec}(\mathbf{V}_1^T\widehat{\mathbf{V}})\|. \tag{93}$$

Using Lemma 5.8, we write $\operatorname{vec}(\mathbf{V}_1^T\widehat{\mathbf{V}}\boldsymbol{\Lambda}) = (\boldsymbol{\Lambda}\otimes\mathbf{I}_{p-d})\operatorname{vec}(\mathbf{V}_1^T\widehat{\mathbf{V}})$, and $\operatorname{vec}(\boldsymbol{\Lambda}_1\mathbf{V}_1^T\widehat{\mathbf{V}}) = (\mathbf{I}_d\otimes\boldsymbol{\Lambda}_1)\operatorname{vec}(\mathbf{V}_1^T\widehat{\mathbf{V}})$; consequently,

$$\begin{aligned}\operatorname{vec}(\mathbf{V}_1^T\widehat{\mathbf{V}}\boldsymbol{\Lambda} - \boldsymbol{\Lambda}_1\mathbf{V}_1^T\widehat{\mathbf{V}}) &= (\boldsymbol{\Lambda}\otimes\mathbf{I}_{p-d})\operatorname{vec}(\mathbf{V}_1^T\widehat{\mathbf{V}}) - (\mathbf{I}_d\otimes\boldsymbol{\Lambda}_1)\operatorname{vec}(\mathbf{V}_1^T\widehat{\mathbf{V}}) \\ &= (\boldsymbol{\Lambda}\otimes\mathbf{I}_{p-d} - \mathbf{I}_d\otimes\boldsymbol{\Lambda}_1)\operatorname{vec}(\mathbf{V}_1^T\widehat{\mathbf{V}}), \end{aligned} \tag{94}$$

and hence,

$$\|\mathbf{V}_1^T\widehat{\mathbf{V}}\boldsymbol{\Lambda} - \boldsymbol{\Lambda}_1\mathbf{V}_1^T\widehat{\mathbf{V}}\|_{\mathrm{F}} = \|(\boldsymbol{\Lambda}\otimes\mathbf{I}_{p-d} - \mathbf{I}_d\otimes\boldsymbol{\Lambda}_1)\operatorname{vec}(\mathbf{V}_1^T\widehat{\mathbf{V}})\|. \tag{95}$$

We now consider the matrix $\boldsymbol{\Lambda}\otimes\mathbf{I}_{p-d} - \mathbf{I}_d\otimes\boldsymbol{\Lambda}_1$. The first term may be written as

$$\boldsymbol{\Lambda}\otimes\mathbf{I}_{p-d} = \operatorname{diag}(\overbrace{\underbrace{\lambda_r,\dots,\lambda_r}_{p-d}, \underbrace{\lambda_{r+1},\dots,\lambda_{r+1}}_{p-d}, \dots, \underbrace{\lambda_s,\dots,\lambda_s}_{p-d}}^{d}), \tag{96}$$

and the second term may be written as:

$$\mathbf{I}_d\otimes\boldsymbol{\Lambda}_1 = \operatorname{diag}(\underbrace{\overbrace{\lambda_1,\dots,\lambda_{r-1},\lambda_{s+1},\dots,\lambda_p}^{p-d}, \overbrace{\lambda_1,\dots,\lambda_{r-1},\lambda_{s+1},\dots,\lambda_p}^{p-d}, \dots, \overbrace{\lambda_1,\dots,\lambda_{r-1},\lambda_{s+1},\dots,\lambda_p}^{p-d}}_{d}) \tag{97}$$

(that is, d copies of $\lambda_1,\dots,\lambda_{r-1},\lambda_{s+1},\dots,\lambda_p$).

Therefore, the matrix $\boldsymbol{\Lambda}\otimes\mathbf{I}_{p-d} - \mathbf{I}_d\otimes\boldsymbol{\Lambda}_1$ is diagonal, with diagonal entries of the form $\lambda_j - \lambda_k$ where $r \le j \le s$ and either $1 \le k \le r-1$ or $s+1 \le k \le p$. The smallest such number, in absolute value, is $\min\{\lambda_{r-1}-\lambda_r, \lambda_s-\lambda_{s+1}\}$; and consequently, continuing (95) we have the following lower bound:

$$\begin{aligned}\|\mathbf{V}_1^T\widehat{\mathbf{V}}\boldsymbol{\Lambda} - \boldsymbol{\Lambda}_1\mathbf{V}_1^T\widehat{\mathbf{V}}\|_{\mathrm{F}} &= \|(\boldsymbol{\Lambda}\otimes\mathbf{I}_{p-d} - \mathbf{I}_d\otimes\boldsymbol{\Lambda}_1)\operatorname{vec}(\mathbf{V}_1^T\widehat{\mathbf{V}})\| \\ &\ge \min\{\lambda_{r-1}-\lambda_r, \lambda_s-\lambda_{s+1}\}\cdot\|\operatorname{vec}(\mathbf{V}_1^T\widehat{\mathbf{V}})\|, \end{aligned} \tag{98}$$

which, from (93), proves Davis-Kahan.

6 Computation via the SVD

In practice, the principal components of a dataset $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ are not computed by forming the covariance $\mathbf{\Sigma}$ and then finding its eigenvectors. Instead, the principal components are evaluated as the singular vectors of the centered, normalized data matrix $\widetilde{\mathbf{X}} = [\mathbf{x}_1 - \boldsymbol{\mu}, \dots, \mathbf{x}_n - \boldsymbol{\mu}]/\sqrt{n}$. Note that by definition, the covariance $\mathbf{\Sigma} = \widetilde{\mathbf{X}}\widetilde{\mathbf{X}}^T$, and so the left singular vectors of $\widetilde{\mathbf{X}}$ are equal to the eigenvectors of $\mathbf{\Sigma}$, which are the PCs. While these two perspectives are mathematically equivalent (that is, identical in infinite precision), computation of the SVD is more numerically stable in the presence of round-off error. This is because the eigenvalues/singular values of $\mathbf{\Sigma}$ are the squares of the singular values of $\widetilde{\mathbf{X}}$, and consequently the condition number of $\mathbf{\Sigma}$ is the square of the condition number of $\widetilde{\mathbf{X}}$. The result is that components with small singular values may be lost.

Let's see this as follows. For simplicity suppose $p \le n$. Suppose the eigenvalues of $\mathbf{\Sigma}$ are $\sigma_1^2 \ge \sigma_2^2 \ge \cdots \ge \sigma_p^2 > 0$, with PCs $\mathbf{u}_1, \dots, \mathbf{u}_p$; and suppose the right singular vectors of $\widetilde{\mathbf{X}}$ are $\mathbf{v}_1, \dots, \mathbf{v}_p$. Then we can write

$$\widetilde{\mathbf{X}} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \cdots + \sigma_p \mathbf{u}_p \mathbf{v}_p^T, \tag{99}$$

and

$$\mathbf{\Sigma} = \sigma_1^2 \mathbf{u}_1 \mathbf{u}_1^T + \cdots + \sigma_p^2 \mathbf{u}_p \mathbf{u}_p^T. \tag{100}$$

Now, suppose that for all $k \ge r+1$, $\epsilon < \sigma_k/\sigma_1 < \sqrt{\epsilon}$, where ϵ is machine precision. Then all the eigenvalues $\sigma_{r+1}^2, \dots, \sigma_p^2$ are less than $\epsilon\sigma_1^2$, and consequently, they are lost to round-off; numerically, $\mathbf{\Sigma}$ appears to be a rank r matrix, and a numerical algorithm for computing eigenvectors can only accurately compute at most the top r PCs. On the other hand $\widetilde{\mathbf{X}}$ is numerically rank p, and so does not encounter the same difficulties.

Furthermore, the computational cost of computing the SVD of $\widetilde{\mathbf{X}}$ is not more than that of forming $\mathbf{\Sigma}$ and computing its eigenvector decomposition. The cost of computing the SVD of $\widetilde{\mathbf{X}}$ is $O(\min\{p^2 n, pn^2\})$. On the other hand, the cost of computing $\mathbf{\Sigma} = \widetilde{\mathbf{X}}\widetilde{\mathbf{X}}^T$ from $\widetilde{\mathbf{X}}$ is $O(p^2 n)$, and then computing its eigendecomposition has cost $O(p^3)$ for a total cost of $O(\max\{p^2 n, p^3\})$. Note that if $p \gg n$, computing the SVD of $\widetilde{\mathbf{X}}$ is significantly cheaper than computing the eigenvalue decomposition of $\widetilde{\mathbf{X}}\widetilde{\mathbf{X}}^T$. For details on numerical computation of the SVD, see, for example, the reference [1].

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